



Temperature prediction for electric vehicles of permanent magnet synchronous motor using robust machine learning tools

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Abstract

Electric vehicles (EVs) are now having a great interest, not only from researchers or manufacturers but from governments and people. Therefore, research and development for this type of vehicle are very important and even reach the point of necessity. The thermal performance of this type of these vehicles is very important and it needs to be studied because it has a great impact on the efficiency of these vehicles entirely. Therefore, this work extracts the heat of the most vital part of these vehicles, which is the electric motor. Now, the most common installed motor for EVs is the permanent magnet synchronous motor (PMSM). Sometimes it is hardly to install sufficient sensors to measure the temperature of the motor accurately, but through measurements of current and by referring to the specifications of the motor, we can expect the temperatures for this engine. For a large number of experiments, all feasible measurements were taken, and the results were saved to later become the data store required for the prediction process. Several machine learning tools have been used to get the best temperature prediction such as extra-tree, bagging k-nearest neighbors (KNN), voting regressor, random forest, and boosting algorithms.

Keywords Temperature prediction · Electric vehicles · Permanent magnet synchronous motor · Machine learning tools

1 Introduction

Due to their environmental friendliness, Electric Vehicles' (EVs) popularity has grown gradually and dramatically. EVs have zero tailpipe emissions and are considered more efficient than internal combustion engines, and have a significant potential for reducing emissions. As a result, the demand for these vehicles has significantly increased (Chapman 2016; Li et al. 2021). To guarantee the effectiveness of these vehicles, it is vital to regularly assess their

performance to identify areas of vulnerability. The thermal performance of electrical motors was analyzed comprehensively by taking into account all-electric motor features such as voltage, current, motor coils, ambient temperature, motor stator size, and other elements that may influence temperature prediction.

The magnetic material used in the rotor of the highly recommended motor in EVs permanent magnet synchronous motors (PMSMs) is an important component in determining motor characteristics (Omar et al. 2019). The power density of PMSMs is growing day by day due to the appearance of new magnetic materials. Simultaneously, the density of losses caused inside the machine is increasing, causing the temperature of a PMSM to fluctuate rapidly. In order to prevent the winding insulator from melting, the temperature of the winding is detected with a sensor in many applications (A and A C 2021). When the temperature rises, the rotor magnets become demagnetized, which harms the machine functioning in EVs. By taking temperature rise into account, an attempt has been made to investigate the rotor magnet demagnetization impact of PMSM. A temperature estimating methodology based on flux observers is utilized. In addition, the predicted temperature-based speed control unit for EV application is proposed in Chen et al. (2022). As a result,

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estimating the temperature in a PMSM has become essential. Two conventional temperature estimating models are employed in many studies. A first-class model makes the use of electrical parameters with temperature-dependent values (Baneira et al. 2015; Czerwinski et al. 2021; Meng and Zhang 2021; Foti et al. 2022). Another type of temperature estimation model is determining the temperature distribution of a PMSM. Whereas a PMSM's losses have nonlinear correlations with both operating conditions (current, voltage, etc.) and temperature, typical temperature estimating methods will have shortcomings.

A nonlinear auto-regressive with exogenous-input (NARX) model is used for temperature estimation of a PMSM. A NARX model is implemented using an artificial neural network (ANN). ANNs are well-known for their ability to solve nonlinear regression problems as developed in Lee and Ha (2020). The load capacity of a PMSM is limited by rotor temperature and high rotor temperature might cause thermal safety issues in the system. As a result, precise prediction of the rotor temperature of an electric vehicle's PMSM is critical for improving motor performance and system operating safety. The heating process and energy flow path of the motor are explored, as well as the heat energy conversion model of the stator and rotor in Zhou et al. (2020a). The dissipative energy conservation of the stator of the motor rotor temperature was used to build the real-time method for predicting rotor temperature. When the system is switched on, the prediction technique of the initial rotor temperature is fitted using the experimental findings. Finally, the test platform was established to confirm the rotor temperature accuracy. The findings demonstrate that the inaccuracy in estimating the temperature of the motor rotor during dynamic running conditions is very small. The study offers a technique for improving the performance and thermal safety of PMSMs used in EVs (Zhou et al. 2020a).

The deep neural network (DNN) model was built for a computer model of PMSM temperature prediction, and the results show that the DNN model can estimate stator winding temperature change, give technical assistance to temperature early warning systems and assure safe operation of PMSMs as proposed in Guo et al. (2020). A fault-tolerant control strategy with a four-leg inverter is used in permanent magnet temperature (PMT) prediction (Tang et al. 2020). The variation rules of the internal magnetic field, losses, and temperature field of the PMSM before and after fault tolerance are obtained. In addition, the purpose of the study is to observe whether the fault-tolerant operating condition will harm the PMSM thermal failure or not. Then, experiments are used to verify the accuracy of the proposed analysis. On the other hand, two efficient models for PMT estimation of dual three-phase (DT)-PMSMs are proposed in Feng et al. (2020). The proposed models are derived by injecting current into a reference frame that is

not involved in torque production. The proposed models can fully investigate the two sets of machine equations to cancel winding resistance and machine inductances by using current injection. To improve estimation performance, inverter nonlinearity is adjusted in the first model and abolished in the second model. The suggested method is computationally efficient and robust to parameter variations, magnetic saturation, and inverter nonlinearity when compared to existing methods. Furthermore, the current injection does not affect the torque production or control performance of the machine. The proposed estimation method is tested on a DT-PMSM in the lab under a variety of operational situations. The PMSM parameters estimation using Extended Kalman Filter (EKF) is presented in Xu et al. (2015) and Al-Gabalawy et al. (2021). The study provided an approach for estimating parameters or temperature variation, which allows researchers to examine and avoid performance degradation by tracking and changing torque observer parameters. The measured speed harmonic and Kalman filter for PMT estimation are investigated in Feng et al. (2019). The PMT estimation is represented in a state-space model based on the derived temperature model, and the Kalman filter is applied to estimate the PMT from the measured speed harmonic. The drive system is examined in the lab under different speed and load conditions. The high-frequency (HF) signal injection technique has been widely examined for PMT estimation (Feng et al. 2017, 2018). It uses the cross-coupling effect for calculating the HF resistance with a linear thermal model then determining the temperature of the PMSM. The stator and rotor temperature estimation based on the energy model for PMSM used in hybrid electric vehicles is presented in Dilshad et al. (2016). The motor losses and a cooling system model are used to determine temperature behavior. The stator and rotor losses are computed individually. The heat transmission between the rotor, stator, and coolant is simulated using convection. The difference between the energy losses and the energy absorbed by the cooling system is used to estimate the temperature variation inside the motor. From all of the aforementioned research studies, it appears that, as yet, there is a little attention has been paid to develop a comprehensive approach for estimating parameters. The contribution of this work can be summarized as follows:

1. Estimate the PMSM temperature based on many supervised machine learning algorithms.
2. Assess the motor performance, where this estimation enables the motor driver the best operation mode, which might be represented in future work.
3. Many sensors have been installed to measure many features such as motor speed, motor torque, air gap temperature, and the temperature of the permanent magnet parts.

4. The collected data has been stored in a data file through the measurement interface card.
5. Many processes have been executed to ensure that all data features are represented correctly.
6. Applying many machine learning tools such as Extra-Tree, Bagging k-nearest neighbors (KNN), Voting Regressor, Random forest, and Boosting Algorithms.

The rest of the paper is organized as follows: In Sect. 2, the system description including mathematical modeling of PMSM and data collection are illustrated. Section 3 presents the main idea of the applied algorithms for the PMSM temperature prediction. In Sect. 4, the results of the applied algorithms for the temperature are discussed. Finally, the conclusions are presented in Sect. 5.

2 System description

2.1 Mathematical modeling of PMSM

According to the PMSM principle, changing input current tends to the production of a changing electric field, which can result in the creation of a rotating magnetic field (B_s). The intensity of the rotating magnetic field depends on the value of the input current. When the three-phase alternating current conducts the stator winding, the rotor's permanent magnet generates a constant magnetic field (B_r). When the B_s generates once powered on, it produces electromagnetic torque, which causes the rotor magnetic field to pursue the stator magnetic field, and it produces a three-phase alternating induction electromotive force (E_0) in the stator winding by the field B_r . The single-phase equivalent circuit of PMSM is illustrated in Fig. 1 (Chaudhari and Fernandes 2001; Elmetwaly et al. 2022).

From Fig. 1, using Kirchhoff's law yields the following equations:

$$U = E_0 + I(R_a + jX_t) \quad (1)$$

$$X_t = X_a + X_\delta \quad (2)$$

where R_a represents the resistance of the stator winding; X_a is the leakage reactance of the stator; X_δ represents the armature reaction reactance; X_t represents the synchronous

reactance; I is the input current; U is the input voltage. The amplitude of the E_0 is proportional to the magnetic flux and the rotation speed of the rotor magnetic field, as stated in Eq. (3).

$$E_0 = K\phi\omega \quad (3)$$

where K represents the motor constant; ϕ is the magnetic flux of the rotor magnetic field; ω is the rotating speed of the motor. As of the constant magnetic flux and known rotating speed, the E_0 can be determined. If the angle between B_s and B_r can be measured, E_0 is acquired. Therefore, Eq. (4) can be stated as follows:

$$U - E_0 = U' = I(R_a + jX_t) \quad (4)$$

2.2 Data collection and processing

The term "data processing cycle" refers to the series of stages or processes that are used to convert raw data into a readable form and create useful information. The growth of various sectors depends on the availability of data, processing, and reliability of data sources. The data processing cycle is shown in Fig. 2.

2.3 Data processing methods

In the Manual data processing method, data is manually processed without the use of any equipment or instrument to get the desired result. Also, all the logical operations and calculations are achieved manually on the data. Similarly, data is transferred manually from one place to another. This technique of data processing is extremely slow, and errors in the result are possible. Many small businesses, as well as government organizations and agencies, handle data manually. Mark sheets, fee receipts, and other financial transactions are all performed manually in an educational institute, for example. This technique is avoided as much as possible due to the significance error, as well as the fact that it is labor-intensive and time-consuming. This kind of data

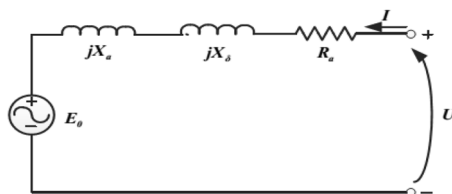


Fig. 1 The equivalent circuit of the PMSM

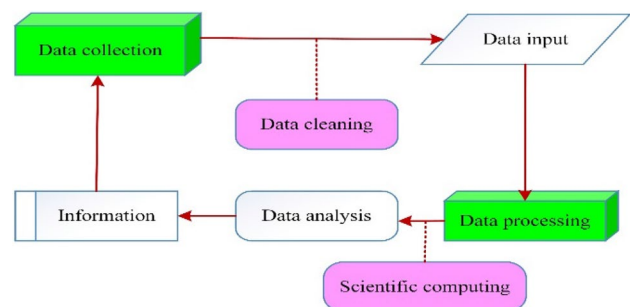


Fig. 2 The data processing cycle

processing dates back to a time when technology was either unavailable or prohibitively expensive. The dependence on manual procedures has lessened dramatically as technology has advanced (Livera et al. 2021; Lafuente et al. 2021).

In the Mechanical data processing method, data is processed by utilizing different devices like mechanical printers, typewriters, or other mechanical devices. This method is more accurate and faster than manual data processing. This form of processing began to fade away with the invention and evolution of more complicated devices with more computing capability. Mechanical data processing machines are often used on examination boards and printers. Electronic data processing (EDP) is a contemporary data processing technology. The data is processed by a computer; data and a set of instructions are entered into the computer, and the computer processes the data automatically following the instructions. An electronic data processing machine is another name for a computer (Yan et al. 2021).

This form of data processing is both quick and precise. At a computerized education system, for example, students' outcomes are prepared using computers; in banks, client accounts are managed or processed using computers, etc.

Batch processing is a way of organizing data by sorting it into groups for more efficient and systematic processing. Online processing is a technique that makes use of Internet connections as well as equipment that is directly connected to a computer. It is mostly used for data recording and research. Real-Time Processing is a mechanism for acquiring and processing information that may respond virtually instantly to diverse inputs. Remote workstations connected to a single large central workstation or server typically use distributed processing. Automated teller machines (ATMs) are a good example of this type of data processing. The online processing technique allows the use of Internet

connections and equipment that is directly connected to a computer. This enables data to be stored in one location yet utilized in another one. Cloud computing is a good example of how this type of processing is used. It is mostly used for data recording and research. The *Real-time processing* method can gather and analyze information nearly instantly in response to various signals. These have a high up-front cost and require a lot of upkeep due to the advanced technology and computer power. Because of output is visible in real-time, the amount of time saved is maximized. Figure 3 shows the stages of the data processing cycle (Chang et al. 2016).

2.4 Complete system description

As well known, the motor temperature is measured to keep the motor performance within efficient limits; as the temperature rises, so does the failure rate. Temperature increases may have an impact on the stator winding and permanent magnet parts. As a result, numerous temperature sensors have been installed in the air gap zone as illustrated in Fig. 4. Figure 5 illustrates the overall configuration of the measurement and data processing where, the measurements were taken using the d-SPACE platform, with temperature sensors connected to interface cards. These data are recorded in a data file. Many data processing methods are applied to this data to eliminate any anomalies or repetitive signals, or to predict the ones that were missed. Subsequently, this data is fed into estimating algorithms like Support Vector Machine (SVM) or Extreme gradient boosting (Xgboost), and the predicted temperature is determined. This temperature is then connected to the motor's control system, and various actions are dependent on temperature values, where increasing the temperature leads to a reduction in necessary

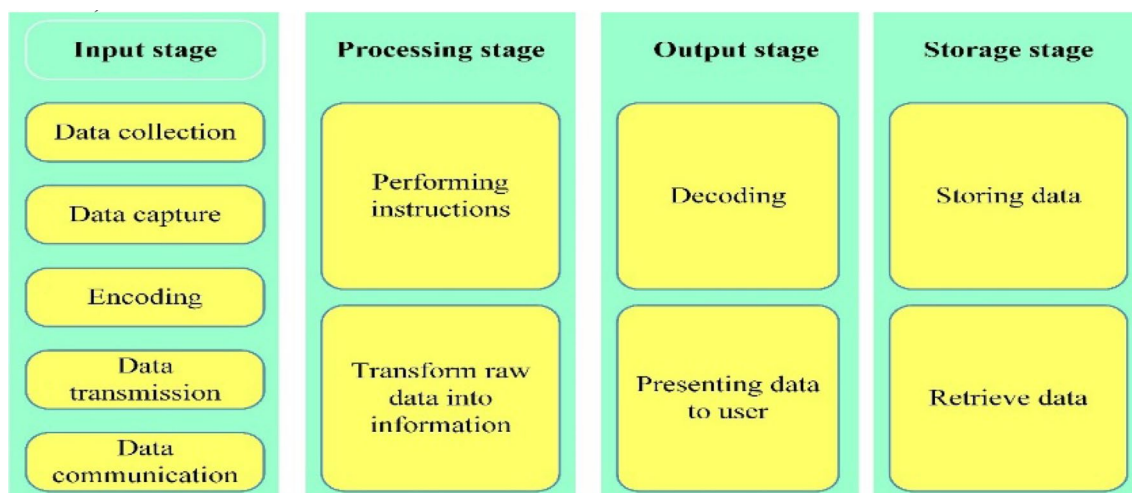


Fig. 3 The stages of the data processing cycle

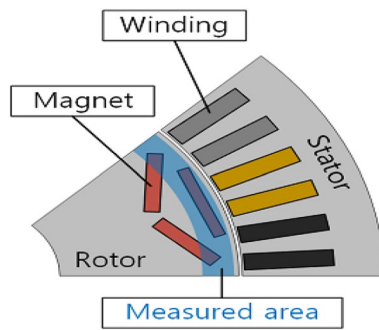


Fig. 4 Measurement zone of the PMSM

developed torque, a decrease in motor speed, or an increase in motor ventilation.

3 Applied algorithms

3.1 Support vector machine

As a result of several of these shortcomings that were stated earlier, a new machine learning technology known as the SVM was created in the last decade as a non-linear solution for classification and regression tasks (Zhou et al. 2020b). At least three factors have contributed to the SVM's effectiveness in producing dependable results: its ability to learn

well with a small number of features, its robustness against model error, and its computing efficiency when compared to other machine learning approaches such as neural networks (Li et al. 2019; Al-Gabalawy 2021a, 2021b). The SVM is commonly categorized into two categories: support vector classification (SVC) and support vector regression (SVR), however, SVR has come to prominence in many petroleum and mining engineering applications. A linear decision function in the SVR-related discussion is defined as described in Eq. (5).

$$f(x) = w^T x + b \quad (5)$$

Using the n -dimensional weighting vector w and the bias parameter b , a vector-like x is utilized to estimate the scalar vector of Y . Although, the main difference between an SVR and traditional regression analysis is that when using an SVR, the decision function is chosen to have the least deviation from the insensitivity parameter ε . This means that the SVR ignores the inaccuracy caused by the data contained inside the ε margins and considers the remainder while determining the ideal hyperplane using slack variables (ξ_i) (Al-Gabalawy 2021b).

As a result, the regression analysis employing the SVR would have an objective function L_p , as shown in Eq. (6), whose goal is to identify the ideal value of the weighting vector w to minimize the empirical risk (Al-Gabalawy 2021b; Rawa et al. 2021). The constraints in Eq. (7) guarantee that

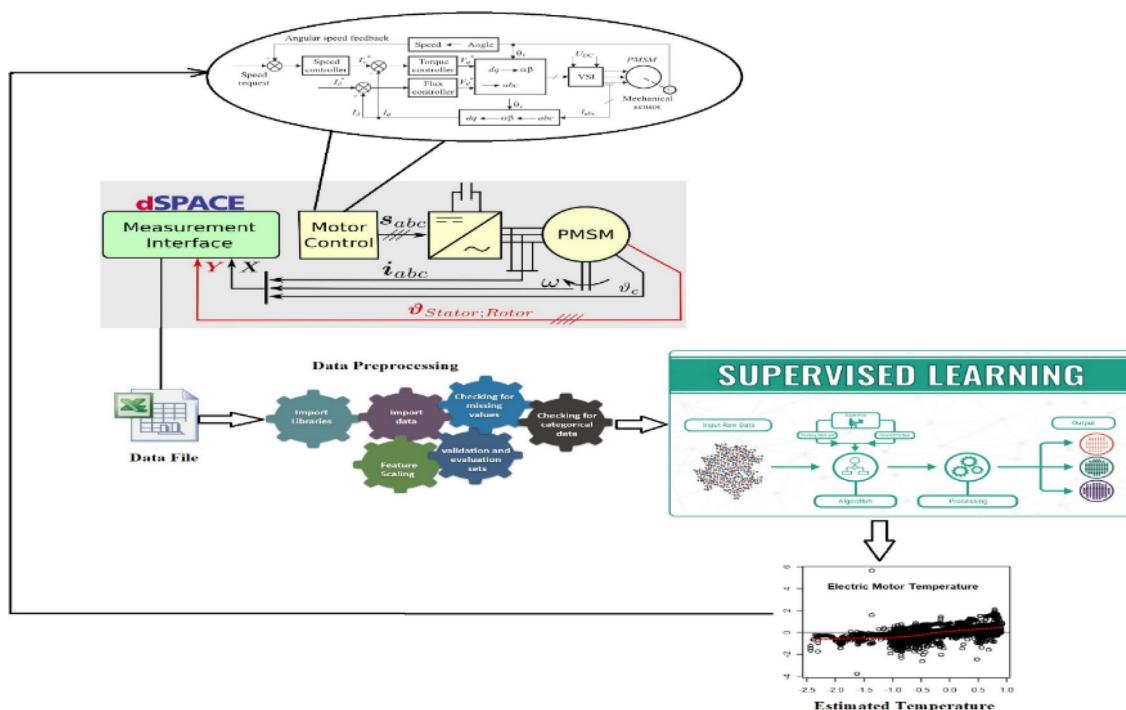


Fig. 5 A holistic configuration of the measurement

any error smaller than ε does not enter the objective function (Omar et al. 2020; Ahmed et al. 2021).

$$L_p = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N (\xi_i + \xi'_i) \quad (6)$$

$$\text{s.t.} \begin{cases} y_i - w^T x - b \leq \xi_i + \varepsilon \\ y_i + w^T x + b \leq \xi'_i + \varepsilon \\ \xi_i, \xi'_i \geq 0 \end{cases} \quad (7)$$

This concept is well known as the Vapnik insensitivity theory (Chen et al. 2020). Same as classification cases, to resolve the optimization problem expressed as Eq. (8), Lagrange multipliers (α_i, α'_i) can be used followed by satisfying the Karush–Kuhn–Tucker (KKT) condition as:

$$\frac{\partial L}{\partial w} = 0 \Rightarrow w = \sum_{i=1}^N (\alpha_i - \alpha'_i) x_i \quad (8)$$

$$\frac{\partial L}{\partial b} = 0 \Rightarrow \sum_{i=1}^N (\alpha_i - \alpha'_i) = 0 \quad (9)$$

Substituting Eqs. (8) and (9) into Eq. (6), the general equation of the linear SVR is formulated as follows in Eq. (11) (Chen et al. 2020):

$$L_d = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N (\alpha_i - \alpha'_i) x_i^T x_j (\alpha_i - \alpha'_j) + \sum_{i=1}^N ((\alpha_i - \alpha'_i) y_i - (\alpha_i + \alpha'_i) \varepsilon) \quad (10)$$

where s.t. in the range of $0 \leq (\alpha_i - \alpha'_i) \leq C$, after locating nonzero Lagrange multipliers, the weighting parameter w of the decision function is calculated using Eq. (8), whereas the bias parameter b is calculated using one of the following equations (Ali et al. 2022):

$$\begin{aligned} -y_i + w^T x_i + b + \varepsilon &= 0 \\ y_i - w^T x_i - b + \varepsilon &= 0 \\ \alpha_i, \alpha_j &< C \end{aligned} \quad (11)$$

The main principle behind developing the SVR for nonlinear data analysis is the same as it was for the linear case, in which an insensitive margin is developed to lower the risk of predictions. However, the problem of mapping data into the high-dimensional Hilbert space must be resolved. The optimal value of the weighting vector w for the regression task may be determined from Eq. (8), which is expressed in Hilbert space as Eq. (12) (Zhang et al. 2020).

$$w = \sum_{i=1}^N (\alpha_i - \alpha'_i) \varphi(x_i) \quad (12)$$

However, as previously stated, the value of parameter φ in the Hilbert space is unknown, so estimating the weighting vector is challenging. In order to solve this problem, apply the kernel trick again, this time substituting Eq. (12) into the decision function. (i.e., $f(x) = w^T \varphi(x_i) + b$) to formulate the general equation of the SVR for the nonlinear data analysis as Eq. (13) (Zhang et al. 2020):

$$\begin{aligned} y_i &= \sum_{i=1}^N \sum_{j=1}^N (\alpha_i - \alpha'_i) \varphi(x_i)^T \varphi(x_j) + b \\ &= \sum_{i=1}^N \sum_{j=1}^N (\alpha_i - \alpha'_i) K(x_i, x_j) + b \end{aligned} \quad (13)$$

The kernel technique eliminates the need to determine the weighting vector, and the bias parameter b may be determined using the Eq. (14):

$$b = y_i - \sum_{i,j=1}^{N_{SV}} \alpha_i y_i K(x_i, x_j) \quad (14)$$

3.2 Extreme gradient boosting

Xgboost is one of the Implementations of Gradient Boosting Machines (GBM), which is known as one of the top-performing algorithms for supervised learning. It may be applied to both regression and classification applications. Therefore, Data scientists choose Xgboost because of its fast execution speed outside of core computation (Luo et al. 2018). The Xgboost operates as follows: For example, suppose we have a dataset DS with m characteristics and n instances.

DS = $\{(x_i, y_i) : i = 1 \dots n, x_i \in \mathbb{R}^m, y_i \in \mathbb{R}\}$. Let $\hat{y}_i$ be the predicted output of an ensemble tree model generated from the Eq. (15).

$$A^{\circ}_{.i} = \phi(x_i) = \sum_{k=1}^N f_k(x_i), f_k \in \mathcal{F} \quad (15)$$

where K denotes the number of trees in the model as shown in Fig. 4, and f_k represents the (k -th tree), we need to determine the optimum set of functions by minimizing the loss and regularization goal to solve Eq. (16).

$$\mathcal{L}(\phi) = \sum_i 1(y_i, A^{\circ}_i) + \sum_k \Omega(f_k) \quad (16)$$

where l is the loss function, which is the difference between the expected and actual output $\hat{y}_i$. While Ω is a measure of the model's complexity, it aids in preventing over-fitting, and it is computed using Eq. (17).

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (17)$$

where T denotes the number of leaves on the tree and w denotes the weight of each leaf. To minimize the objective function in decision trees, boosting is employed in training the model, which works by adding a new function as the model continues to train. As a result, on the t -th iteration, the following new functions (tree) in Eq. (18–21) are then added (Li et al. 2018b, a, 2019):

$$\mathcal{L}^{(t)} = \sum_{i=1}^n 1(y_i, A^\circ A_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (18)$$

$$\mathcal{L}_{\text{split}} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma \quad (19)$$

$$g_i = \partial_{A^{(t-1)}} l(y_i, A^{(t-1)}) \quad (20)$$

$$h_i = \partial_{A^{(t-1)}}^2 l(y_i, A^{(t-1)}) \quad (21)$$

4 Results and discussions

The model's evolution is shown in the second part, which includes both linear and nonlinear models. It explains how the model was created and exhibits the model at various stages along the development process. The code notebook (Python 3) investigates additional models that can suit the data well after assessing the linear regression models. The completed model, as well as an analysis and interpretation of the model, is subsequently provided. The results of applying the model to estimate home prices using the data in the development dataset are shown in this section. The paper then goes through how to use Ridge and Lasso Regression to add penalty terms to a model and improve its accuracy. Lasso Regularization may also be used for feature engineering, as it shows us which factors have the most impact on the target variable, which is crucial for the Inference job. Finally, the text emphasizes non-linear models as a mean of refining the model and displays the Root Mean Square Error (RMSE) number. Because the visual output of the Q-Q plot and the residual vs fitted plot indicated a nonlinear connection, nonlinear models were used. The last part

contains information on the model metrics to be compared and the model that was chosen. There are 15,147 instances in the data collection, each with 13 columns. The first eight represent the model properties, which are; ambient, coolant, u_d , u_q , motor speed, motor torque, i_d , and i_q . The remind features are; rotor temperature (rotor temperature is noted as pm), stator yoke, stator winding, and stator tooth. And it is found that the stator yoke, stator winding, and stator tooth did not change through the measurement, so there is no need to consider them in the study.

4.1 Prediction task

For the prediction task, the underlying problem is to predict the rotor temperature using the collected 8 attributes. There are many steps are required to measure the model performance. The first step is to divide the raw data into two sets: training and testing (profile ids 72 and 81). Fit the model with the training data, make predictions using the test data, and calculate performance measures. At least one model that differs from my final model is being compared. Defining and defending the models that were chosen. The final phase is to evaluate, analyses, and interpret the findings.

The main objective of this work is to determine which critical elements have a significant impact on rotor temperature. Variable correlation analysis and regression equations can be used to make the required inferences. Statistical evidence such as correlations and p-values, as well as interpret the selected attribute subset to demonstrate the influence of numerous factors. As shown in Table 1, the first quartile (Q1) indicates that 25% of observations are below this number, while the third quartile (Q3) indicates that 75% of observations are below this quantity.

Divide the original data into two sets: training and testing. Only the data associated with profile ids 72 and 81 should be included in the test data set, while the remainder should be kept in the training data. Table 2 shows that the train dataset comprises 9652 rows and 8 columns, whereas the test dataset has 5495 rows and 9 columns

4.2 Exploratory data analysis

Exploratory data analysis (EDA) is a great approach to illustrate the properties of the data and the insights present in the data. univariate, bivariate, and multivariate analysis have been applied to explore the data, and Fig. 6 reveals that outliers exist in the eight variables; ambient, coolant, u_d , u_q , motor speed, motor torque, i_d , i_q and the target variable pm .

Outliers may be quite important, and merely removing them will have a major influence. However, none of these outliers will be deleted since we do not have enough information about the range of these variables to determine whether an outlier is due to a data mistake or a genuine value. Many

Table 1 Variable correlation analysis

Attribute	Min	Q1	Median	Mean	Q3	Max	No of unique values
Ambient	-3.328	-0.539	-0.140	-0.107	0.498	2.711	14,263
Coolant	-1.237	-0.691	-0.045	0.181	1.010	2.295	14,513
u_d	-1.654	-0.486	0.300	0.110	0.350	2.273	15,126
u_q	-1.815	-1.268	-0.381	-0.191	0.741	1.770	15,100
Motor speed	-1.222	-1.222	-0.432	-0.191	0.674	2.024	9434
Motor torque	-3.316	-0.258	-0.256	-0.084	0.357	3.014	9788
i_d	-3.176	-0.552	0.587	0.177	1.029	1.060	10,484
i_q	-3.309	-0.246	-0.246	-0.081	0.350	2.914	12,451
pm	-2.605	-0.811	-0.120	-0.127	0.515	2.905	15,139
Stator yoke	-1.8278	-0.7347	-0.0574	0.0462	0.8692	2.4363	13,973
Stator tooth	-2.0571	-0.9005	-0.0697	-0.0408	0.7786	2.2887	13,656
Stator winding	-2.0006	-0.9607	-0.0925	-0.0897	0.6249	2.5499	14,758

Table 2 The train dataset

#	Ambient	Coolant	u_d	u_q	Motor speed	Motor torque	i_d	i_q
1	0.1177	-1.041	-1.0267	0.9251	1.2125	0.3587	-0.8343	0.3229
2	-0.5548	-1.087	-1.2259	0.5126	2.0241	0.2842	-1.3638	0.2340
3	-0.7722	-1.064	0.2989	0.9562	-0.1402	-0.2591	1.0291	-0.2457
4	0.2200	-1.083	-1.0261	0.9278	1.2125	0.3574	-0.8381	0.3228
5	0.4016	-1.079	0.3155	-0.7094	-0.9519	-0.2563	1.0292	-0.2457
6	-0.7698	-1.090	0.2999	0.9590	-0.1402	-0.2593	1.0291	-0.2457
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
9654	-0.1435	-0.55149	0.2990	-1.2715	-1.2224	-0.2556	1.0291	-0.2457
9655	-0.3561	1.74732	1.3822	1.2475	0.8691	-0.8936	-0.1888	-0.8834
9656	-0.1591	-0.72018	0.3498	-1.2639	-1.2224	-0.2556	1.0291	-0.2457
9657	-0.1303	-0.09399	-1.1096	0.4486	-0.1825	1.2612	0.1768	1.3878
9658	-0.1411	-0.83836	1.0281	1.4850	0.9783	-0.6664	-0.2155	-0.6495

of the variables have negative values, which explains why log conversions are difficult and certain variables, such as i_q , motor torque, motor speed, and coolant have median values less than 0, implying that many values are less than 0. Outliers don't affect medians, which is why they're so useful. There are no not applicable values in any of the variables, indicating that the dataset contains no null values. The skew of a perfect normal distribution is 0. Ambient has a skewness of -0.64 and i_d has a skewness of -0.83, indicating a left-skewed distribution with a strong negative skew. Coolant has a +0.46 skew, u_q has a +0.41 skew, and motor speed has a +0.53 skew. These variables all have a fairly positive skew, indicating that they are skewed to the right. The variables pm and u_d have practically 0% skew, thus they don't need to be transformed. Torque and i_q have a tiny positive skew (right-skewed), thus no modification is required.

Density plots are used to investigate the distribution of one or a few variables, as shown in Fig. 7. When you

obtain a new dataset, the first thing you should do is check the distribution of your variables one by one. It provides a substantial amount of data. The mean and median of a completely symmetrical distribution are the same. We will examine the numerical properties of ambient, coolant, u_d , u_q , motor speed, torque, i_d , and i_q , as well as our objective variable, pm .

The histogram for the characteristics is shown in Fig. 8. A histogram is a chart that depicts the distribution of a numeric variable's values as a series of bars. A bar's height reflects the frequency of data points having a value within the relevant bin; each bar generally spans a range of numeric values called a bin or class.

In the bivariate analysis, where the pairs of the variables are investigated. In other words, a plot could get to show the performance of two variables together, for example, Fig. 9a shows that the pm is increasing as the values of ambient increase but coolant does not show such a consistent linear

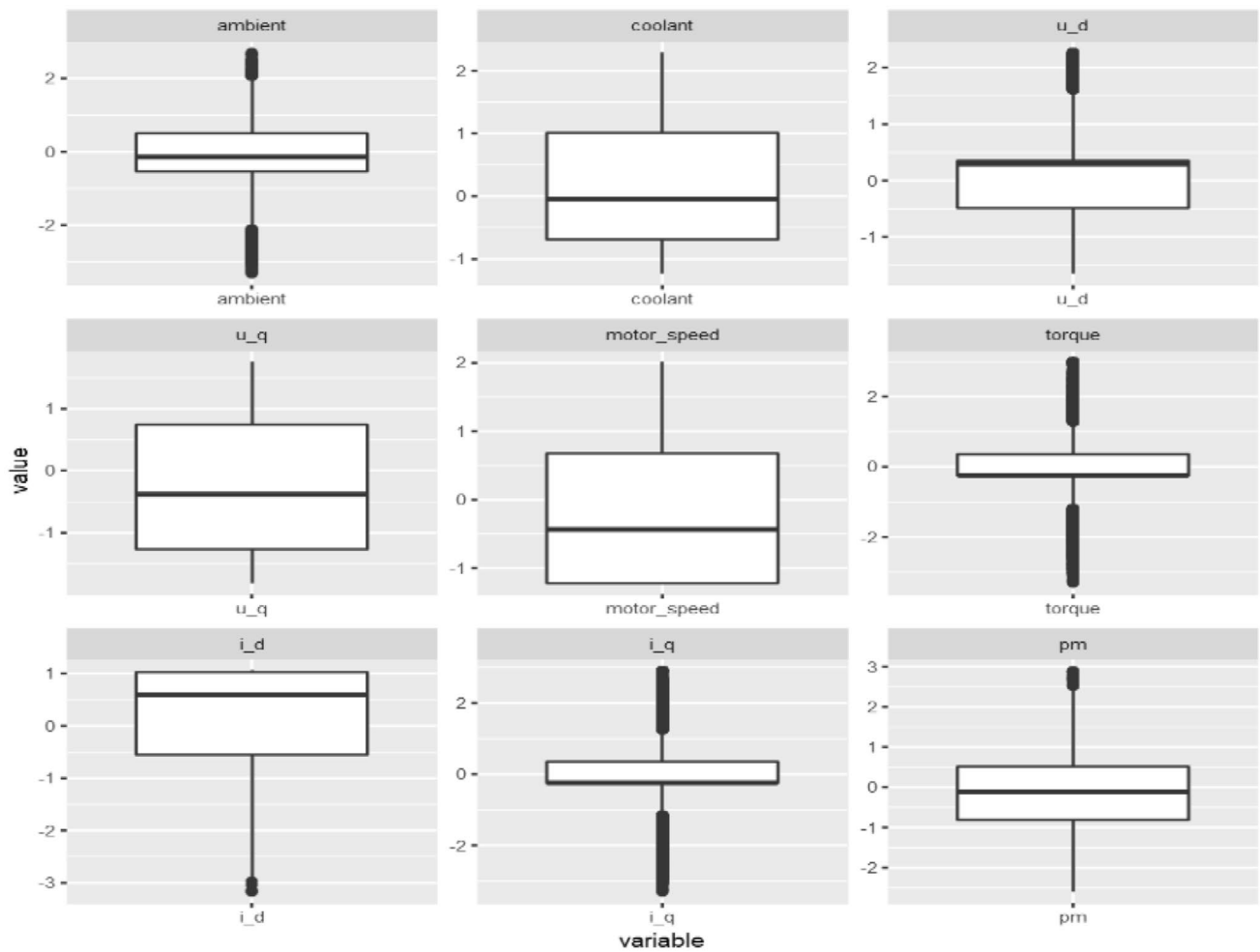


Fig. 6 Exploratory data analysis

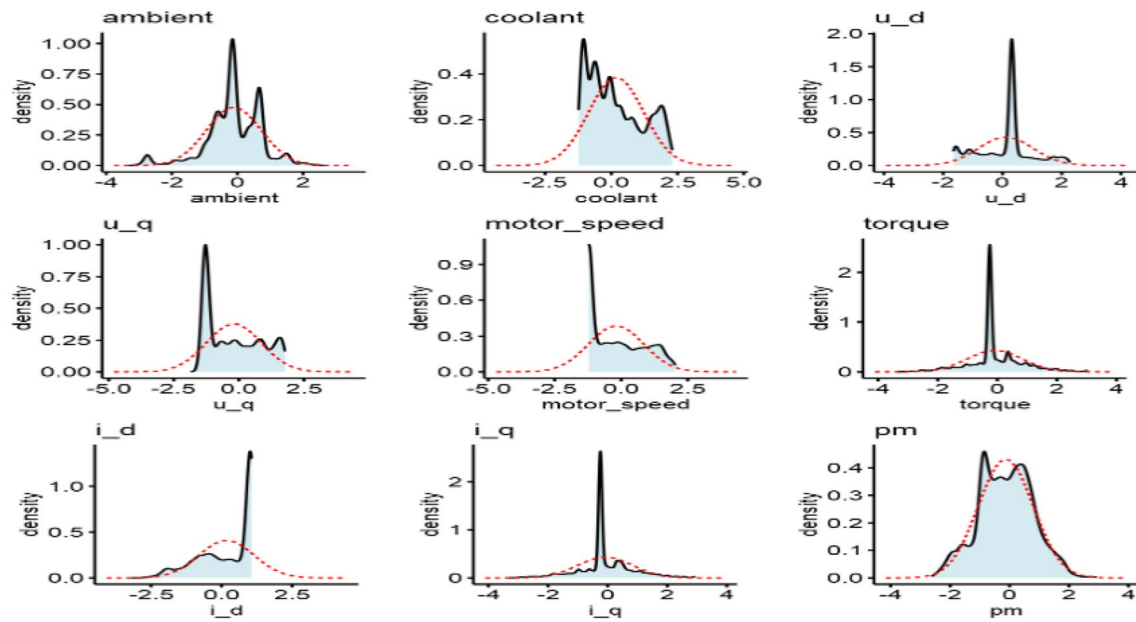


Fig. 7 Density plots

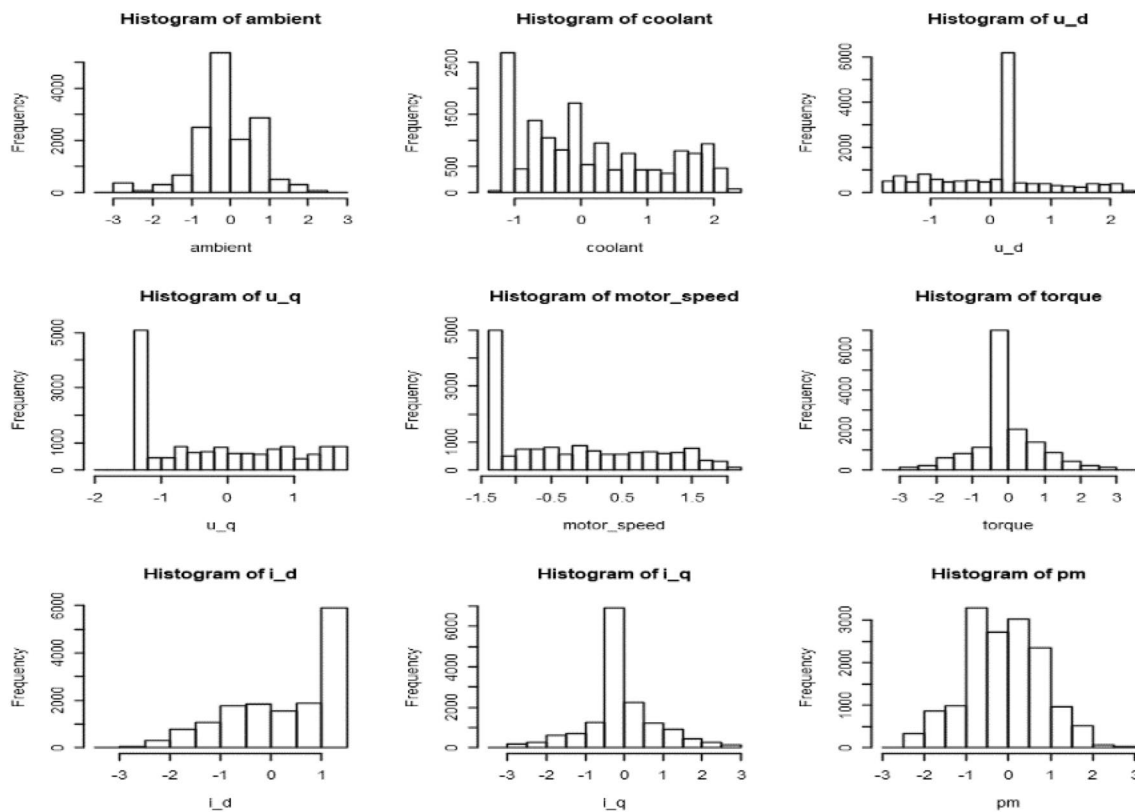
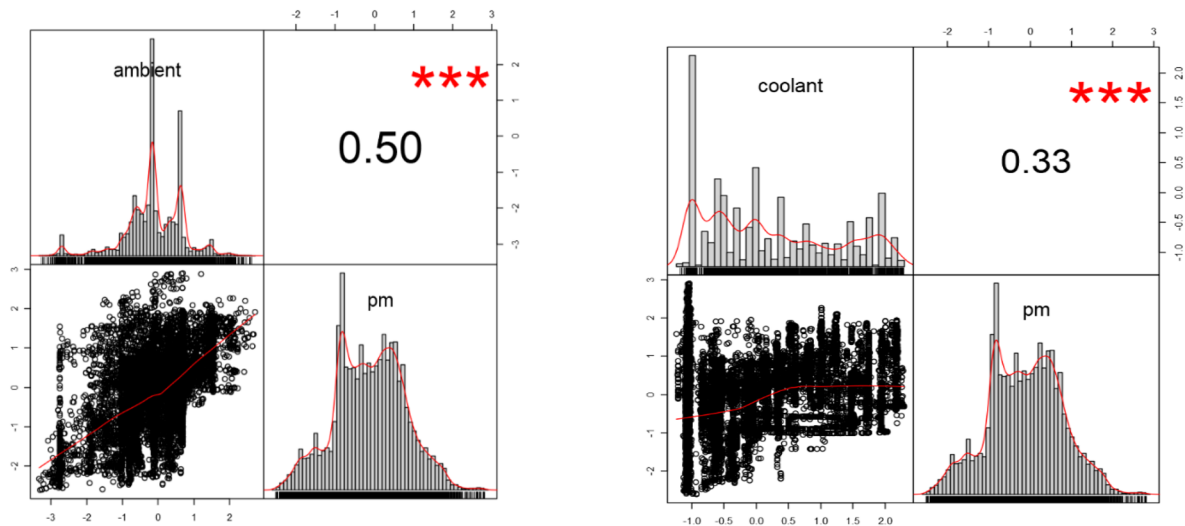


Fig. 8 The distribution of a numeric variable's values



(a) The bivariate analysis between the ambient and the motor temperature.

(b) The bivariate analysis between the coolant and the motor temperature.

Fig. 9 the bivariate analysis between the selected features. (a) The bivariate analysis between the ambient and the motor temperature. (b) The bivariate analysis between the coolant and the motor temperature

association. On the other side, Fig. 9b that relation between coolant and pm , shows that the values are increasing and slowly stabilizing after a point. Finally, both ambient and

coolant are positively correlated with the target variable pm . While, in the multivariate analysis, a correlation matrix could be used to show the correlation between the different

features, wherein Fig. 10, it is easy to note that u_d strongly negatively correlated with torque and i_q . The correlation between u_d and i_d is moderate. There is a correlation between u_q and motor speed, while there is a highly positive correlation between torque and i_q . Finally, it could be noted that there is a decent correlation between pm , ambient, coolant, motor speed, and i_d . From the following correlation matrix, Table 3 shows the association between ambient with pm is a correlation coefficient of 0.50. Association between coolant with pm is a correlation coefficient of 0.33 ambient and coolant have the highest correlation with pm among all of the variables.

4.3 Comparative analysis

Linear regression model, polynomial regression, SVM, and Xgboost have been applied to estimate the motor temperature. The linear regression is used to check if the data would be fit using linear relationships or not. If not, SVM

and Xgboost are implemented to deal with non-linear data. Regularizations techniques L_1 and L_2 are applied to reduce the model complexity, where L_1 refers to a lasso regression while L_2 refers to ridge regression. Traditional cross-validation techniques using tenfold cross-validation (CV) are used to better m model the data.

The linear regression model is a basic model that may be used as a base model to determine whether or not there is a linear connection between the dependent (goal) and independent variables using statistical metrics, residuals, and other approaches mentioned in the model development session. Linear Regression is used for this dataset because there are multiple independent variables, and the formula is presented as in Eq. (22). While polynomial regression is applied to overcome under-fitting and to convert the original features into their higher-order terms we will use the Polynomial Features. This function fits a polynomial regression model to powers of a single predictor by the method of linear least squares as in Eq. (23).

$$Y_i = \beta_0 + \beta_1 X_{i,1} + \dots + \beta_p X_{i,p} + \varepsilon_i, \text{ and } i = 1, \dots, n \quad (22)$$

$$Y_i = \beta_0 + \beta_1 X_{i,1}^2 + \dots + \beta_p X_{i,p}^2 + \varepsilon_i, \text{ and } i = 1, \dots, n \quad (23)$$

where Y represents the dependent variable, X shows the independent variables, ε is defined as an error variable, and β_i describes the regression coefficients, while β_0 is an intercept term. Therefore, in this work, X here is the eight selected attributes, while Y is the motor temperature.

Ridge regression is based on the same assumptions as linear regression: linearity, constant variance, and independence. Ridge regression is a model tuning approach for analyzing any data that have multicollinearity, with an extra penalty parameter that seeks to reduce complexity. L_2 regularization is performed using this approach. When there is a problem with multicollinearity, least-squares are unbiased, and variances are big, the projected values are distant from the actual values.

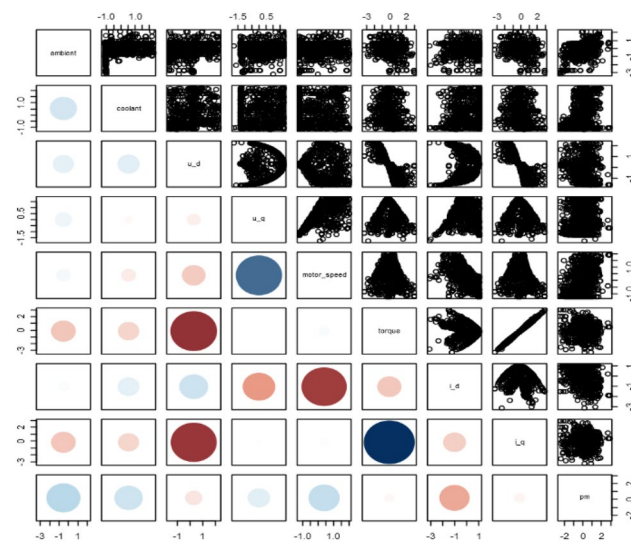


Fig. 10 Multivariate analysis

Table 3 A correlation matrix

	Ambient	Coolant	u_d	u_q	Motor speed	Motor torque	i_d	i_q	pm
Ambient	1.00000	0.29806	0.14523	0.090443	0.09340	-0.21692	0.02151	-0.21526	0.49579
Coolant	0.29806	1.00000	0.15828	-0.04001	-0.0698	-0.161076	0.13281	-0.157715	0.33273
u_d	0.14523	0.15828	1.00000	-0.04424	-0.1880	-0.83460	0.31703	-0.81129	-0.0809
u_q	0.09044	-0.0400	-0.0442	1.000000	0.79804	-0.00530	-0.3623	0.001397	0.13218
Motor speed	0.09340	-0.0699	-0.1879	0.798042	1.00000	0.033811	-0.7707	0.019384	0.31229
Motor torque	-0.2169	-0.1611	-0.8346	-0.00530	0.03381	1.000000	-0.2285	0.996858	-0.0548
i_d	-0.0215	0.13281	0.31703	-0.36226	-0.7707	-0.22852	1.00000	-0.20009	-0.2886
i_q	-0.2153	-0.1577	-0.8113	0.001397	0.01938	0.996858	-0.2001	1.000000	-0.0652
pm	0.49579	0.33273	-0.0809	0.132183	0.31229	-0.05482	-0.2886	-0.06524	1.00000

Regularization is a crucial concept for avoiding data overfitting, especially when the learned and test data differ significantly. Regularization is done by adding a "penalty" term to the best fit produced from the training data in order to achieve a lower variance with the tested data, as well as compressing the coefficients of predictor variables to limit their effect on the output variable. The least absolute shrinkage and selection operator is a type of regularization. For a more accurate forecast, it is preferred over regression approaches. Shrinkage is used in this model. As shown in Eq. (24), shrinkage occurs when data values are shrunk towards a central point known as the mean.

$CostFunction + \lambda *$

(Sum of the absolute value of the magnitude of coefficient)
(24)

Xgboost is a gradient boosting-based decision-tree-based ensemble Machine Learning technique. Xgboost uses parallel processing and tree pruning to handle regularization and increase model performance. It is a great blend of software and hardware optimization approaches that produce higher outcomes in the shortest period while utilizing fewer computer resources.

The gradient boosting machine employs a boosting mechanism to help poor learners become stronger. It teaches a model to predict value by reducing mean square error (MSE), $\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$, where $\hat{y} = F(x)$ is the predicted value, y is the actual value, and n is the number of samples. The gradient boosting has been executed for M stages, and the weakest model F_m can only estimate the mean of y for each stage (m), ($1 \leq m \leq M$). The equation of the line in Support Vector Regression is $y = wx + b$, which is comparable to linear regression. This straight line is referred to as hyperplane in Support Vector. The nearest data points on either side of the hyperplane are termed support vectors, and they are used to plot the boundary line. The SVR, unlike other regression models, aims to fit the best line inside a threshold value, rather than minimizing the error between the real and projected value. The goal is to think about the points that fall within the decision boundary line. A kernel aids in the search for a hyperplane in higher-dimensional space while reducing the computing cost.

The RMSE is a standard deviation of prediction error(residual) which is the difference between actual value (y) and predicted value ($\hat{y}$). It evaluates how to spread the predicted value is from the actual value, thus, the lower RMSE the better the prediction.

The RMSE, as in Eq. (25), is the standard deviation of the prediction error (residual), which is the difference between the actual (y) and expected values ($\hat{y}$). It assesses

how far off the expected and actual values are; the smaller the RMSE, the better the prediction.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (25)$$

One of the most used statistical measurements is R-squared, as shown in Eq. (26). It refers to the fluctuation in the dependent (target) variable caused by the independent variable.

$$R^2 = 1 - \frac{RSS}{TSS} \quad (26)$$

where RSS is $\sum_{i=1}^n (\hat{y}_i - y_i)^2$, and TSS equals $\sum_{i=1}^n (y_i - \bar{y})^2$. So, Eq. (26) is converted to Eq. (27).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (27)$$

Therefore, the Residual standard error and F-statistics are calculated as $\sqrt{\frac{1}{df} RSS}$ and $\frac{(TSS-RSS)/p}{RSS/df}$ respectively. The basic multiple linear regression model is utilized initially, followed by a stepwise function to eliminate the least influential variables and maintain just the most relevant ones. I calculated that R-squared is just 0.491 based on the above model. In this model, the F-statistic is $1.17e+03$. At the 0.05 level, the p -values for the coefficients demonstrate that ambient, coolant, u_d , u_q , motor speed and torque are significant. Similarly, the t value for these variables is substantially higher and distant from 1. This model explains 49.1 percent of the fluctuation in electric rotor temperature, according to the corrected R-squared value. The F-statistic $1.38e+03$ has a p -value of $2e-16$, indicating that the model is useful and rejecting the null hypothesis.

Figure 11 shows that stepwise doesn't boost performance significantly because all the plots are the same; it only helps with the model's complexity. The Residuals versus Fitted plot, as well as the Normal Q-Q plot, show that the tail is significantly distant from the fitted line, implying a non-linear connection. As a result, the remarks may be summed up as follows: Residual versus Fitted—plot suggests non-linear connection as The error terms fluctuate along the regression line, hence this model violates the homoscedasticity condition. Normal Q-Q—the residuals do not depart greatly from the dashed line, but they do shift towards the conclusion of the plot, showing that the residuals are more or less normally distributed but not at the end. The values larger than 2.0, on the other hand, deviate from the dashed line. We can also see that there are relatively few probable outliers, such as sample number 1909. Fitted Values/Scale-Location Plot—The graph demonstrates that the model largely adheres to

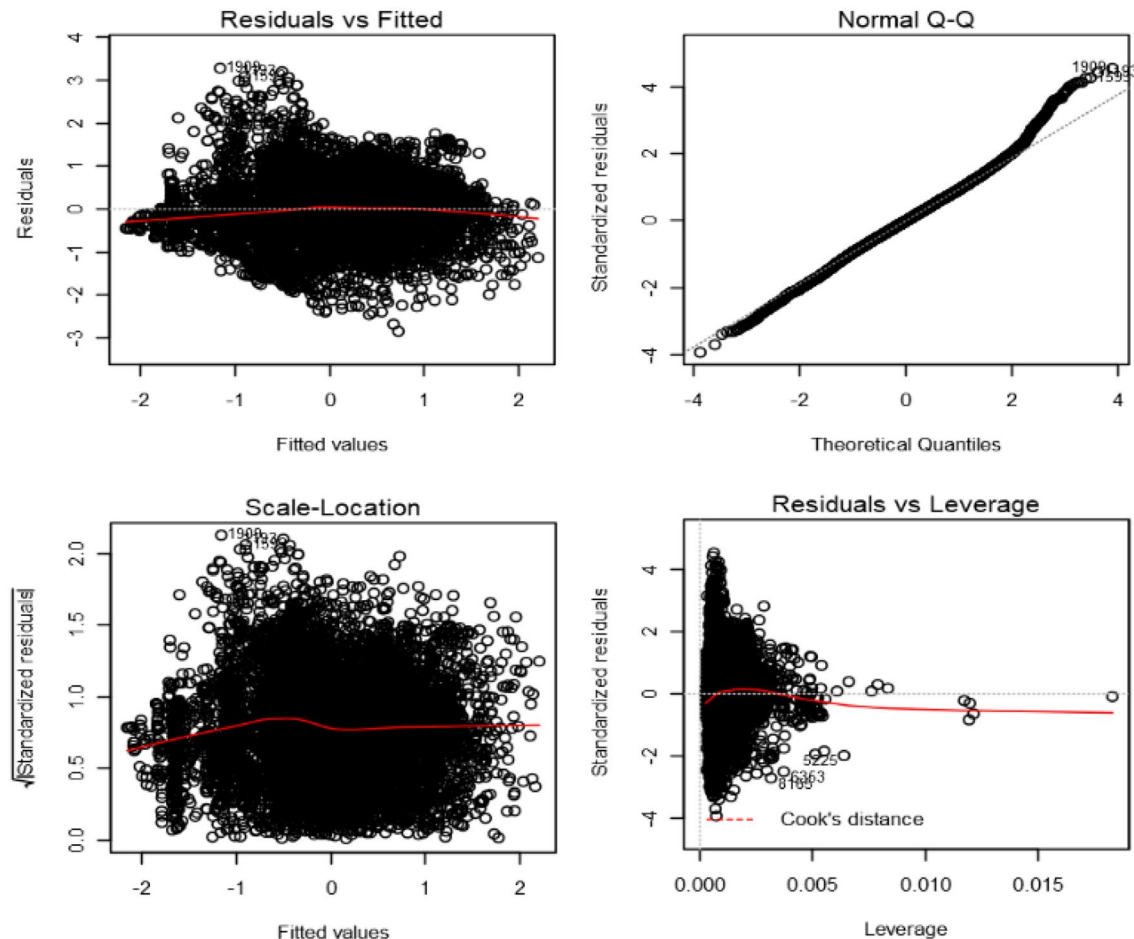


Fig. 11 Residual versus fitted, normal Q-Q, scale location, and residual versus leverage

the assumption of equal variance, while it does so marginally in the case of fitted values (-2 , -1). Residuals vs. Leverage—It aids in the identification of data's influential samples, such as outliers, leverage samples, and influential observations. It also aids in the detection of heteroscedasticity and nonlinearity.

There are some potentially influential outliers, as seen in Fig. 12. The step function appears to identify the factors that have a greater influence on the target variable. Running backward in time hasn't eliminated any variables, so they're all significant. In this exercise, we will add the interaction terms to the model and see if it helps or not. Determine which variables are highly associated and add some interaction terms for the correlated variables. When an independent variable has a varying influence on the target variable based on the values of its independent variables, this is referred to be an interaction term. The influence of interaction terms on the target variable is typically relevant in multiple linear regression models. The squared-R value has increased from 0.42 to 0.55, and the residual plots remain in good shape. The graphs illustrate that there

are a few potentially important outliers that might have an impact on the value of it.

Ridge regression is a model tuning technique that may be used to analyze data with multicollinearity. L_2 regularization is performed through the Ridge regression. When there is a problem with multicollinearity, least-squares are unbiased, and variances are big, the projected values are distant from the actual values. In this job, it is used model tuning and selection approaches like Ridge and Lasso Regression to increase the accuracy of the model over the prior linear regression model. Repeated tenfold cross-validation would be used, which is a resampling procedure whose goal is to test a machine learning model's ability to predict new data to identify problems such as overfitting or selection bias, as well as to provide insight into how the model will generalize to a different dataset. As in Fig. 13a, the coefficients will be penalized in the ridge regression, so those that are the least efficient in your estimation will decrease the fastest. I'm punishing more and lowering model complexity as a boost Lambda. Ridge regression has increased the test RMSE by a small

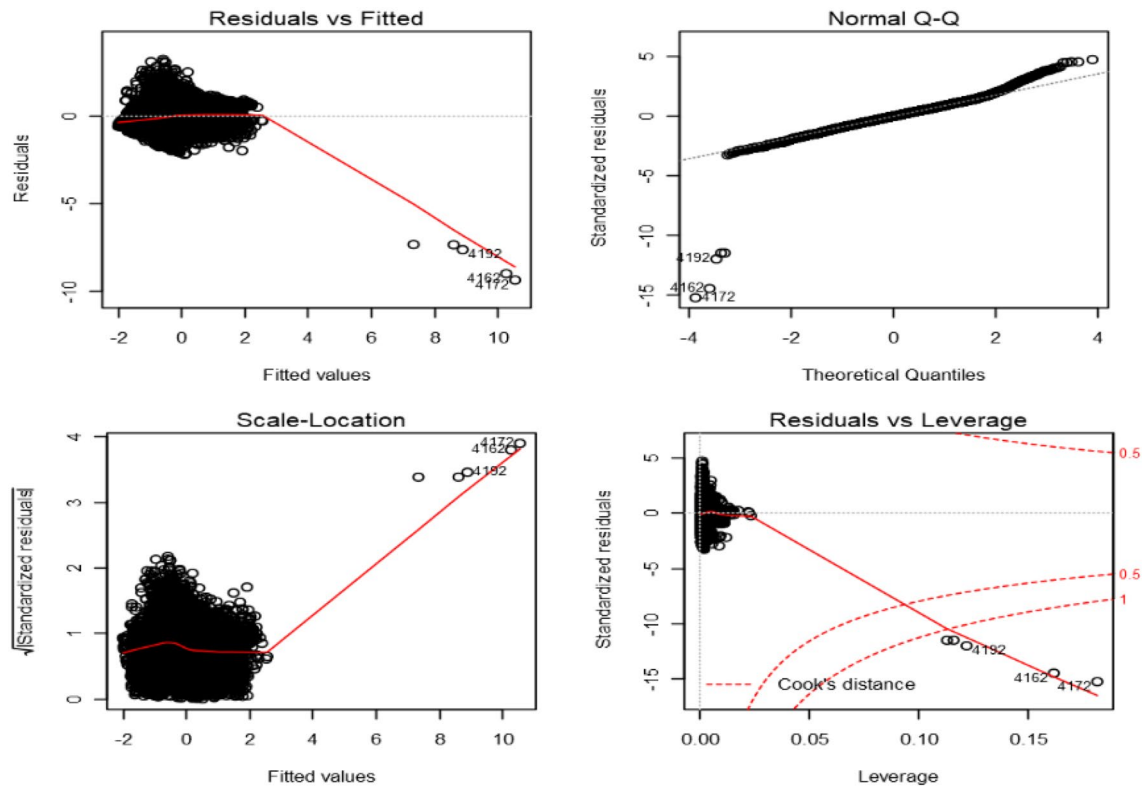


Fig. 12 Some potentially influential outliers

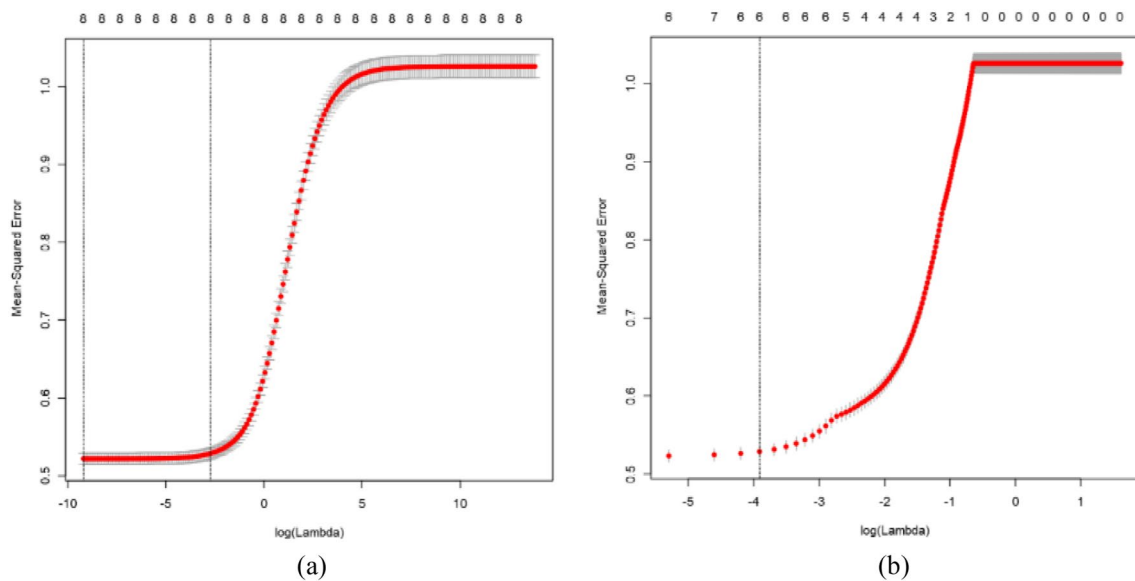


Fig. 13 Coefficients in the ridge regression

amount, to 0.7095, which is a substantial improvement over the prior model.

As a result, we may conclude that including a penalty element decreases model complexity while increasing accuracy. While Lasso regression is used to find the subset of

predictors that minimizes the target variable's prediction error. It applies a penalty term to the model parameters called lambda, which causes the coefficients for the variables to decrease toward zero as in Fig. 13b. This aids in the selection of the most important model parameters. As

a result, it may be utilized for feature engineering as well. The outcomes are similar to those of basic linear and ridge regression. It is only a little affected by Lasso Regression.

There are certain crucial places if Polynomial Regression raised to the power 2 is used. I'd also try to get rid of the outliers and then double-check the model's accuracy to determine if Polynomial Regression models generalize the testing data successfully. As in Fig. 14, the R2 value has increased from 0.42 for the linear regression model (base model) to 0.593 for the polynomial regression model, and the residual plots have improved, with some potentially relevant outliers. After eliminating outliers, rebuild the model.

Due to the non-linear relationship between the predictor variables and the target response, linear models are unable to generalize successfully. SVR and Xgboost are used to create two non-linear models. Xgboost, a variation of Gradient Boosting, has been used in this work. *Xgboost* is a gradient boosting-based decision-tree-based ensemble machine learning technique. The hyperparameters have been adjusted, where the parameter C, which is shared by all SVM kernels, trades off the misclassification of training instances for the decision surface's simplicity. A low C smooths the decision surface, whereas a high C attempts to properly anticipate all training examples. The importance matrix should be demonstrated as in Table 4, where the

Table 4 The importance matrix

Feature	Gain	Cover	Frequency	Importance
Coolant	0.30655	0.18511	0.18511	0.30655
Ambient	0.21617	0.11199	0.16733	0.21617
Motor speed	0.13116	0.11507	0.09799	0.13116
u_d	0.12480	0.13827	0.14232	0.12480
u_q	0.07045	0.15424	0.14575	0.07045
Motor torque	0.05811	0.09118	0.08495	0.05811
i_d	0.05724	0.12095	0.11527	0.05724
i_q	0.03552	0.08688	0.06127	0.03552

Gain denotes the relative contribution of the associated feature to the model, which is determined by adding the contributions of each feature to each tree in the model. When compared to another attribute, a greater value for this metric indicates that it is more relevant for creating a prediction. The Cover metric refers to the number of observations that have been made about this feature. This will be computed for all features, and the cover will be 17 as a percentage of the total cover metrics for all four features. The Frequency is a percentage that represents the number of times a specific feature appears in the model's trees.

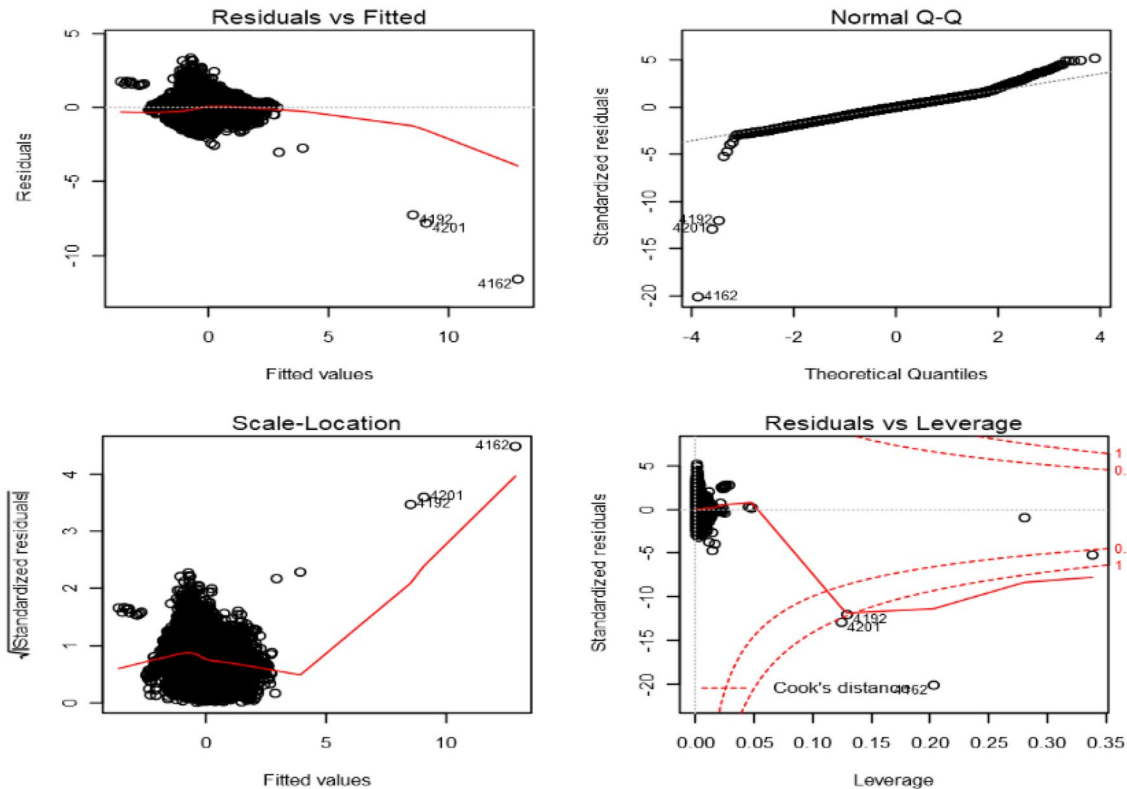


Fig. 14 Values of R2 in linear and polynomial regression

The multiple Linear Regression has been applied and a step function to exclude all of the other variables with less significance from the linear models. It was extremely difficult to fit the linear model since the data had nonlinear interactions. As a result, the model's R -Squared=0.491 accuracy is low, but the RMSE is high, with RMSE Train=0.72 and RMSE test=0.709. The model's accuracy is improved by the interaction terms. Ridge Regression: Ridge regression attempts to minimize model complexity by adding a penalty term to the model parameters. The accuracy R -Squared=0.571 appears to be better than the prior model, however, the RMSE train=0.7219 and RMSE test=0.7095 indicate no significant increase as in Table 5. To decrease model complexity and comprehend the most significant elements of the model, lasso regression also seeks to add an extra penalty term to the model parameters. However, there has been no discernible improvement.

Due to the data being nonlinear, nonlinear models are applied, which gives a good performance. With RMSE train = 1.226 and RMSE test = 0.8, Xgboost with hyperparameter adjustment performs much better. In tree-based techniques, R -Squared is not a valid metric of model comparison. We may conclude that the model is well generalized based on the test RMSE. We can also observe from the variable significance matrix that ambient, coolant, and motor speed are the three most important predictors for the model. SVM performs a similar function as *Xgboost*. RMSE Train = 1.2 and RMSE Test = 0.5 are the values. This nonlinear model is also an excellent data model. We can also observe from the variable significance matrix that ambient, coolant, and motor speed are the three most important predictors for the model. SVM is a supervised machine learning tree-based decision tree, while SVR is a regression model version of SVM. We can also observe from Fig. 15 that the nonlinear model fits the data pretty well.

5 Conclusions

This study focuses on using exploratory data analysis (EDA) and predictive models to estimate permanent magnet motor temperature. The first task focuses on the distribution of

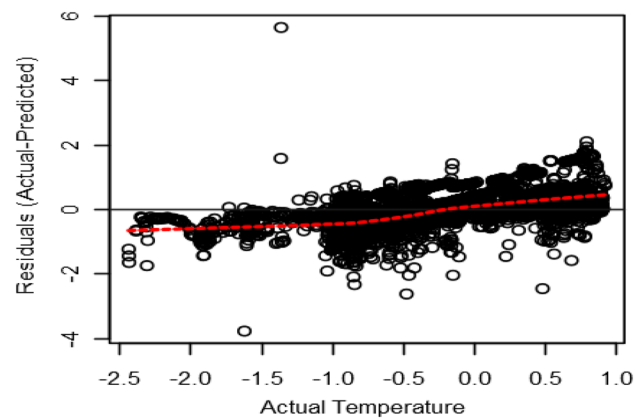


Fig. 15 Residual versus actual temperature

each attribute and their relationships to one another from EDA; later, models such as multiple linear regression model, polynomial regression, Xgboost, and SVM will be trained on the Train dataset, and evaluation metrics such as RMSE, and R -squared will be used to assess the models' performance on the testing and training datasets. In the Results and Discussion section, the models are well compared as well as the variable relevance. After all of the analyses and model comparisons, the SVM is the best model, with the most relevant variables being ambient, coolant, and motor speed. The Test RMSE produced by this model is 0.5, which gives the lowest of all the models. As a result, SVM has a features of: effective in high dimensional spaces, effective in cases where number of dimensions is greater than the number of samples, and uses a subset of training points in the decision function (called support vectors), so it is also memory efficient.

In contrast, the Linear Model performs poorly since our dependent and independent variables appear to have non-linear connections. The link between the two variables is determined by fitting a linear equation to observe data in linear regression. One variable is regarded as an explanatory variable, while the other is regarded as a dependent variable. Root Mean Square Error is the error function utilized by the fitted line in this model. For the model to work successfully,

Table 5 Comparative analysis

Model	RMSE train	RMSE test	R_squared
Linear regression	0.729	0.7096	0.491
LM with interaction terms	0.557	0.750	0.551
Ridge regression	0.7219	0.7095	0.601
Lasso regression	0.7219	0.7096	0.601
Polynomial regression	0.75	0.801	0.593
Polynomial regression with no outliers	0.76	0.801	0.682
Xgboost	1.226	0.8291	–
SVM	1.226	0.5898	–

it also demands that the predictor variables be approximately normally distributed. There are various assumptions about homoscedasticity and model error that must be met when using Linear Regression. The error terms are normally distributed and the residuals have constant variance and the same values across all independent variables. It should be observed that the linear model does not appear to match the data well because the dataset has no linearity.

This nonlinearity may also be seen in the Q–Q plot and the Residual vs Fitted plot. As a result, the Decision Tree-based nonlinear model, which in our instance attempts to fit the model effectively and has the lowest RMSE value for the test dataset, is the winner. We may conclude that SVM is a well generalized model for our dataset because it has the lowest RMSE of all the models tested. The Test RMSE is 0.5, indicating that the model has been generalized well. For nonlinear models, the variable importance is also computed to comprehend the measure of impact exerted by the predictor variables. For SVM, it can be seen that ambient, coolant, and motor speed features have the most influence on the PM variable, and the same is true for the Xgboost model. The goal of the inference task is to find the essential elements that have a significant impact on the rotor temperature. This temperature has the most impact for model's performance. For the Inference task, it may be deduced that the most significant influences on rotor temperature are ambient, motor speed, and coolant. For the future, these algorithms might be applied to other topics related to power and energy application such as the estimation of the microgrid power consumption, the generated power from the PV and wind turbine systems.

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